# ASSIGNMENT -2

# K. R. MANGALAM UNIVERSITY

**BTech CSE (AI & ML)**
**Section - B**
**SOET (School of Engineering and Technology)**


**NAME: Nikhil**
**ROLL NO: 2301730077**


**COURSE: Computer Networks**
**Course Code: (ENCS352)**


K. R. MANGALAM UNIVERSITY,
SOHNA ROAD GURUGRAM, HARYANA

## Assignment 2: Packet Flow Visualization Using Simulation Mode

### Experiment Objectives

Students will use Packet Tracer Simulation Mode to observe how packets move between devices, how switches learn MAC addresses, and how ICMP echo request/reply travels through a small network.

### Learning Outcomes

- Use Simulation Mode effectively
- Observe ARP + ICMP behavior
- Understand switch MAC learning behavior
- Interpret packet event list timeline

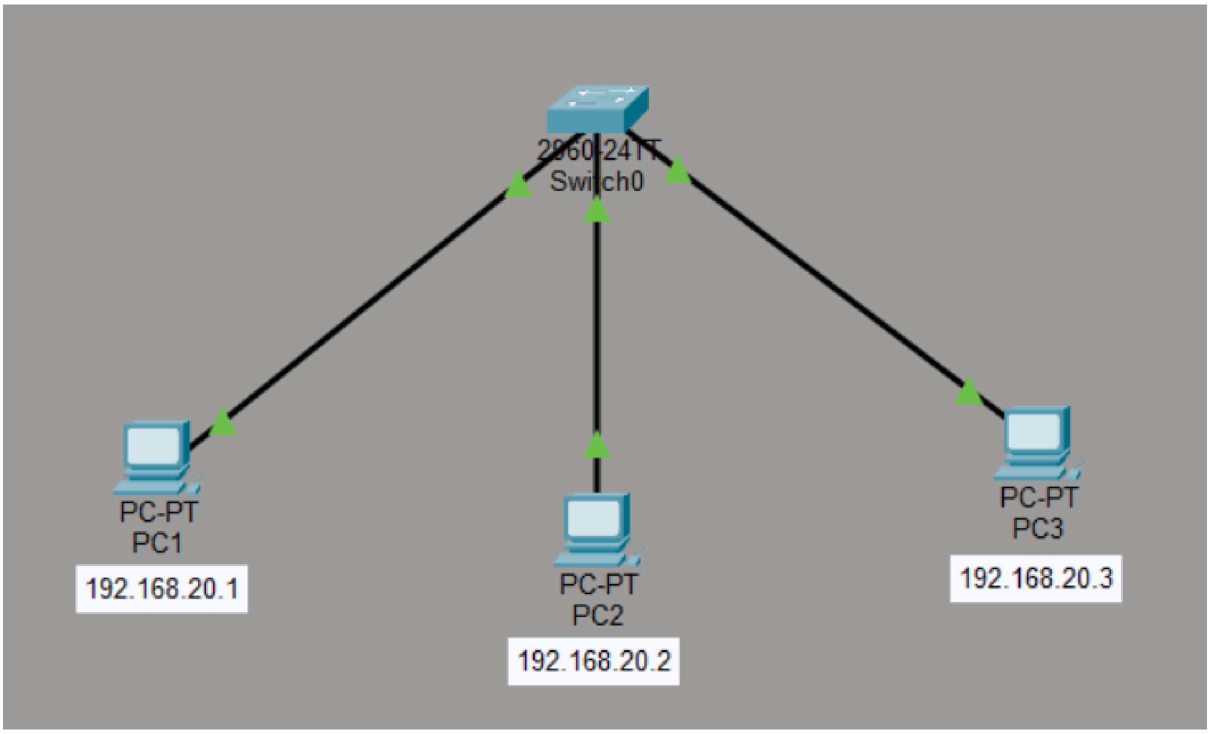
### Concepts Used

- Simulation Mode
- ICMP ping
- ARP resolution
- Switch MAC address table learning

GITHUB LINK: Nik848/CN_LAB_Assignments_KRMU

## Task 1: Build a Simple LAN

- 1 switch + 3 PCs
- IP example: 192.168.20.1–3 /24

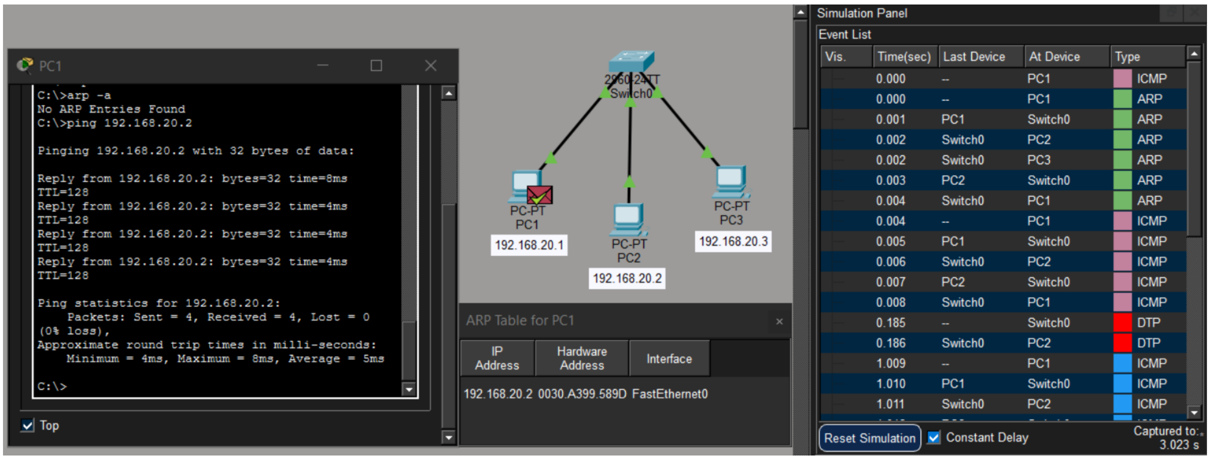



**Observation**

A simple LAN network was created using one switch and three PCs. Each PC was connected to the switch using Ethernet cables and assigned IP addresses in the same subnet (192.168.20.1–192.168.20.3 with subnet mask 255.255.255.0). The network topology was successfully established, allowing the devices to communicate with each other through the switch.

## Task 2: First Ping Observation

- Enable Simulation Mode
- Ping PC1 → PC2
- Observe ARP request and ARP reply before ICMP

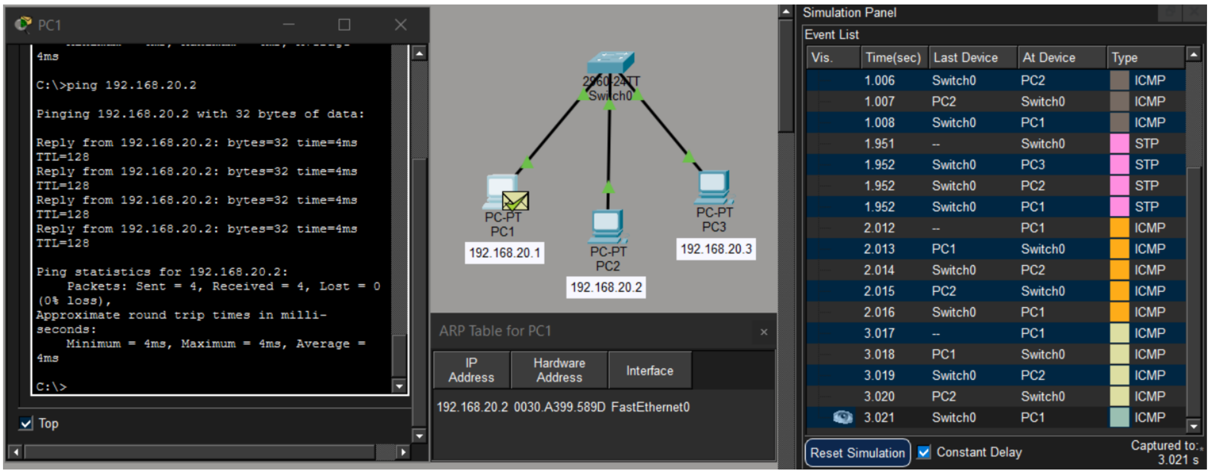



### Observation

During the first ping from PC1 to PC2 in Simulation Mode, an ARP request was broadcast by PC1 to discover the MAC address of PC2. PC2 responded with an ARP reply containing its MAC address. After the MAC address was learned and stored in the ARP table, ICMP echo request and echo reply packets were exchanged successfully between the two devices.

## Task 3: Second Ping Observation

- Ping PC1 → PC2 again
- Observe fewer packets (ARP cached)

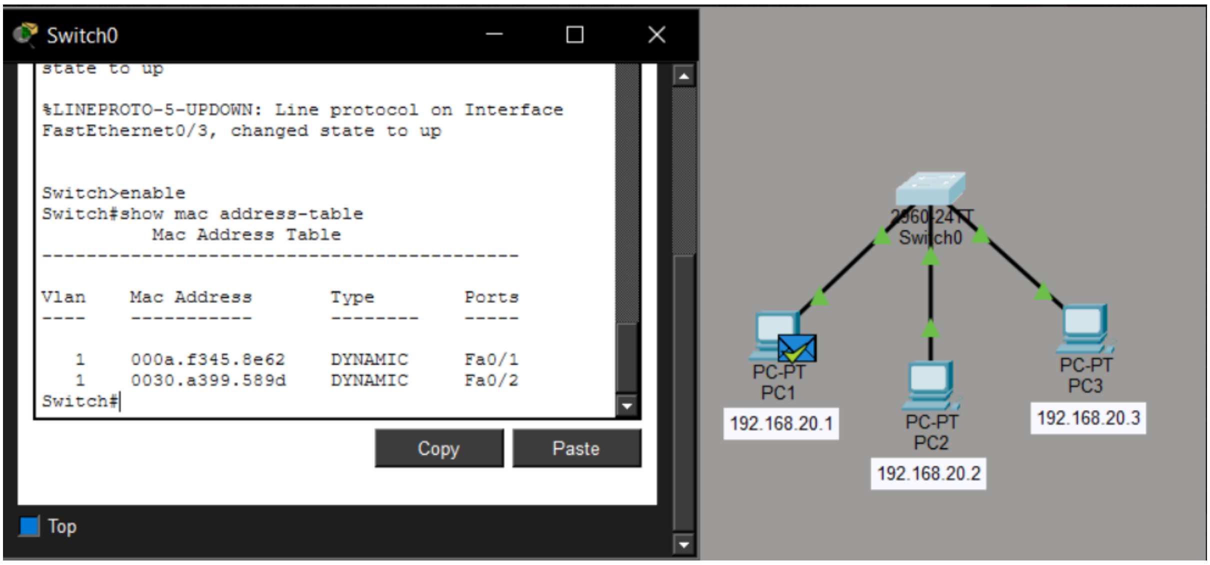



### Observation

During the second ping from PC1 to PC2, only ICMP echo request and echo reply packets were observed in the Simulation Event List. No ARP request or ARP reply was generated because the MAC address of PC2 was already stored in the ARP cache of PC1. This reduced the number of packets exchanged and made the communication faster.

## Task 4: MAC Table Check

- View switch MAC table (if available) / observe switch learning through events




**Observation**

The MAC address table of the switch was checked using the show mac address-table command in the switch CLI. The table displays the MAC addresses of connected devices along with the ports through which they are connected. These entries are dynamically learned by the switch when it receives frames from the devices, allowing it to forward data efficiently to the correct port.

## Short Notes: First Ping vs Second Ping

**First Ping:**

- Before sending data, the source device does not know the destination MAC address.
- An **ARP request** is broadcast to find the MAC address of the destination IP.
- The destination device sends an **ARP reply** with its MAC address.
- After that, **ICMP echo request and echo reply** packets are exchanged.
- Therefore, the first ping takes slightly more time and shows **ARP + ICMP packets**.

**Second Ping:**

- The destination MAC address is already stored in the **ARP cache**.
- No ARP request or ARP reply is needed.
- Only **ICMP echo request and ICMP echo reply** packets are sent.
- Communication is faster and involves **fewer packets**.